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GEOGRAPHY · Natural hazards and disasters and their management

UNDERSTAND IT FIRST

What is it? A **hazard** is a natural event that **can** cause harm (a cyclone at sea, a moving fault). It becomes a **disaster** only when it hits **people who cannot cope** and causes heavy loss of life, property or livelihood beyond the community's capacity. So **Risk = Hazard × Vulnerability ÷ Capacity**: the same cyclone kills thousands where there are no shelters and very few where there is warning and evacuation (compare **Odisha 1999** with **Phailin 2013**). **Disaster management** is everything done **before, during and after** to reduce that risk. Learn it as **India's exposure → each hazard (cause, area, warning) → the management cycle → law and institutions → global frameworks → AP's experience**.

1. India's exposure and the main hazards

- India is one of the most disaster-prone countries. Commonly quoted figures (NDMA): about **59%** of the land is prone to **moderate to very high earthquakes**; over **40 million hectares** (about **12%**) to **floods**; about **5,700 km** of the roughly **7,500 km** coastline to **cyclones and tsunamis**; about **68%** of cultivable land to **drought**; and hilly areas to **landslides and avalanches**. Hazards are **natural** (geological, hydro-meteorological, biological) or **man-made** (industrial, chemical, nuclear, fire, stampedes).

Hazard	Cause and where	Warning and response
Cyclones	Low-pressure systems over **warm seas (above about 26–27°C)**; the **Bay of Bengal** gets about **four times** as many as the Arabian Sea because it is warmer and gets remnants of Pacific typhoons; peaks **Oct–Dec** and **May**; damage from **wind, storm surge** and rain; Odisha, AP, TN, West Bengal most exposed; Gujarat on the west.	**IMD** (Regional Specialised Meteorological Centre, New Delhi) tracks and **names** cyclones for the north Indian Ocean (list of **13 countries**, 2020); colour-coded alerts; **National Cyclone Risk Mitigation Project**: shelters, embankments, warning systems; mangroves as natural barriers.
Floods	Heavy monsoon rain, **silted** rivers, **Himalayan** rivers changing course (Kosi), cloudbursts, dam releases, **encroachment** of floodplains; **Assam, Bihar, UP, West Bengal** most; **urban floods** (Mumbai 2005, Chennai 2015, Vijayawada 2024) from blocked drains and built-over lakes.	**Central Water Commission (CWC)** gives flood forecasts; embankments, reservoirs, **flood-plain zoning**, river linking debates; **Rashtriya Barh Ayog (1976)** assessed flood-prone area.
Drought	A **slow-onset** disaster: **meteorological** (rain deficit), **hydrological** (low water in rivers, tanks, groundwater), **agricultural** (crop failure); **Rajasthan, Gujarat, rain-shadow Deccan, Rayalaseema** (Anantapur).	**Drought Manual (2016, updated 2020)**; watershed works, **MGNREGA**, crop insurance (**PMFBY**), drought-tolerant seeds, micro-irrigation.

Earthquakes and tsunamis	Movement of the **Indian plate** into the Eurasian plate; **Himalaya, NE, Kutch, Andamans** highest risk (e.g. **Bhuj 2001**, **Latur 1993** in the 'stable' Deccan). **Tsunami**: an undersea earthquake pushes up the sea (**26 December 2004**, Sumatra; TN, A&N, Kerala, AP hit).	**BIS seismic zone map** (zones; a revised code in **2025** made the risk map stricter for the Himalaya: check the latest); **earthquake-resistant building codes**; **INCOIS, Hyderabad** runs the **Indian Tsunami Early Warning Centre (2007)**; **National Center for Seismology** reports quakes.
Landslides, heat, lightning	**Landslides**: steep slopes, heavy rain, deforestation, road cutting; **Himalaya, Western Ghats, NE** (**Kedarnath 2013**, **Wayanad July 2024**, Joshimath subsidence 2023). **Heatwaves**: IMD declares when the maximum is at least **40°C** in the plains and **4.5°C or more above normal** (or **45°C** or more actual). **Lightning** kills more Indians than any other natural hazard in many years.	**Geological Survey of India (GSI)** is the **nodal agency for landslides**; **Heat Action Plans** (first: **Ahmedabad 2013**); lightning alerts via **Damini** app and SMS.

2. Managing disasters

The disaster management cycle

Disaster management institutions

- **Law and plans:** after the **Odisha super cyclone (1999)**, the **Gujarat earthquake (2001)** and the **2004 tsunami**, India moved from **relief** to **risk reduction**: the **Disaster Management Act, 2005** created **NDMA** (chaired by the **Prime Minister**), **SDMAs** (**Chief Ministers**), **DDMAs** (**Collector**; the elected zilla parishad chairperson is co-chair), the **NDRF** (raised **2006**, under the **Ministry of Home Affairs**; battalions drawn from the central armed police forces) and **NIDM**. The **National Policy on Disaster Management (2009)** and the first **National Disaster Management Plan (2016**, revised 2019) followed. Money: **National Disaster Response Fund** and **State Disaster Response Funds** (Centre pays **75%**, or **90%** for NE and Himalayan States), plus mitigation funds since the **15th Finance Commission**.
- **Disaster Management (Amendment) Act, 2025:** gives the NDMA and SDMAs the direct job of preparing the national and State plans; creates **Urban Disaster Management Authorities** for State capitals and large cities with municipal corporations; makes the **State Disaster Response Force** statutory; and sets up **national and State disaster databases**. **Global frameworks:** the **Yokohama Strategy (1994)** → **Hyogo Framework (2005–2015)** → **Sendai Framework (2015–2030)**: **4 priorities** (understand risk; strengthen risk governance; invest in resilience; better preparedness and 'build back better') and **7 global targets** (fewer deaths, fewer people affected, lower losses, less damage to infrastructure, more national strategies, more cooperation, better early warning). India's PM gave a **10-point agenda** on disaster risk reduction (2016) and set up the **CDRI (2019)**.

- **Andhra Pradesh:** the **APSDMA** runs a State Emergency Operations Centre with **early warnings by SMS** for cyclones, floods and **lightning** (lightning kills many farm workers in AP); hundreds of **multi-purpose cyclone shelters** along the coast; lessons from **Diviseema (1977)**, **Hudhud (2014)**, **Titli (2018)**, **Michaung (2023)**, **Montha (2025)**; the **Krishna and Budameru floods in Vijayawada (September 2024)**; **heatwaves** in Rayalaseema and Palnadu; **drought** in Anantapur; the **LG Polymers gas leak (Visakhapatnam, May 2020)** as a man-made disaster.

OTHER ANGLES ASKED IN EXAMS

Other angles asked: (1) **Slow-onset vs rapid-onset**: drought and desertification are slow; earthquakes, cyclones, flash floods are rapid. (2) **Storm surge**, not wind, causes most cyclone deaths. (3) **Cyclone names** are given in turn from the list; a used name is never repeated. (4) **'Disaster' is not in the Seventh Schedule** as a separate entry; the DM Act was passed using **Entry 23 of the Concurrent List** (social security and social insurance). (5) **Ministry of Home Affairs** is the nodal ministry for most disasters, but **Agriculture** for drought and **Health** for epidemics. (6) **Odisha** is often cited as a model for cyclone preparedness (zero-casualty approach).

EXAM TRAP

Risk = hazard × vulnerability ÷ capacity; exposure: **~59% earthquake, ~12% (40 m ha) flood, ~5,700 km coast cyclone, ~68% cultivable drought**; **cyclones: warm sea >26–27°C; Bay ~4× Arabian Sea; Oct–Dec, May; storm surge; IMD RSMC New Delhi names (13 countries, 2020); NCRMP**; **floods: CWC forecasts; Rashtriya Barh Ayog 1976; urban floods (Mumbai 2005, Chennai 2015, Vijayawada 2024)**; **drought: slow-onset; meteorological, hydrological, agricultural; Manual 2016**; **earthquakes: Bhuj 2001, Latur 1993; BIS zones (2025 revision)**; **tsunami 26 Dec 2004; INCOIS Hyderabad (ITEWC 2007)**; **landslides: GSI nodal; Kedarnath 2013, Wayanad 2024**; **heatwave ≥40°C + 4.5°C above normal, or ≥45°C; Ahmedabad HAP 2013**; **lightning: Damini**; cycle: **mitigation, preparedness, response, recovery**; **DM Act 2005: NDMA (PM), SDMA (CM), DDMA (Collector), NDRF 2006 (MHA), NIDM**; **NPDM 2009, NDMP 2016**; **NDRF/SDRF funds (75%, 90%)**; **2025 Amendment: UDMAs, statutory SDRF, databases, NDMA/SDMA plans**; **Yokohama 1994 → Hyogo 2005–15 → Sendai 2015–30 (4 priorities, 7 targets)**; **PM's 10-point agenda 2016; CDRI 2019**; **Concurrent List entry 23**; AP: **APSDMA, cyclone shelters, lightning SMS, Diviseema 1977 … Montha 2025, Vijayawada 2024, LG Polymers 2020**.

Test yourself: 8 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Disaster Management Act, 2005:
1. The NDMA is chaired by the Prime Minister.
2. The State Disaster Management Authority is chaired by the Chief Minister.
3. The District Disaster Management Authority is chaired by the District Collector.
4. The NDRF functions under the Ministry of Defence.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. The NDRF works under the Ministry of Home Affairs.

PRACTICE

Match List-I (agency) with List-II (role):

List-I (agency)	List-II (role)
A. INCOIS	**1.** Nodal agency for landslides
B. IMD	**2.** Training and capacity building in disaster management
C. NIDM	**3.** Cyclone tracking and naming for the north Indian Ocean
D. Geological Survey of India	**4.** Tsunami early warning

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) INCOIS: tsunami warning; IMD: cyclones; NIDM: training; GSI: landslides.

PRACTICE

Which one of the following statements is NOT correct?

A. The Sendai Framework covers the period 2005–2015.

B. The Sendai Framework has seven global targets.

C. India's first National Disaster Management Plan was released in 2016.

D. For most States, the Centre provides 75% of the State Disaster Response Fund.

Answer: A. Sendai covers 2015–2030; 2005–2015 was the Hyogo Framework.

PRACTICE

Assertion (A): India's east coast is hit by far more cyclones than the west coast.
Reason (R): The Arabian Sea is warmer than the Bay of Bengal.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the Bay of Bengal is generally warmer and also receives remnants of Pacific storms, which is why it has more cyclones.

PRACTICE

The IMD warns that a severe cyclone will cross the coast near a fishing village tonight with a storm surge of 2–3 metres. What is the safest action for the villagers?

A. Stay inside their low-lying houses with doors locked

B. Go to the beach to secure their boats during the storm

C. Shelter in a basement or underground room

D. Move early to the designated cyclone shelter on higher ground

Answer: D. A storm surge floods low-lying areas, so early evacuation to a raised cyclone shelter saves the most lives.

PRACTICE

Consider the following statements:
I. Drought is a rapid-onset disaster.
II. With present science, the exact time and place of an earthquake cannot be predicted.
Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: B. Drought is a slow-onset disaster that builds over months.

PRACTICE

Consider the following pairs (disaster – year):

1. Kedarnath flash flood – 2013

2. Cyclone Hudhud landfall at Chennai – 2014

3. Bhuj earthquake – 2004

4. Wayanad landslides – 2018

How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: A. Only pair 1 is correct. Hudhud struck Visakhapatnam (2014); Bhuj was 2001; the Wayanad landslides were in July 2024.

PRACTICE

Arrange the following in chronological order:

1. Release of India's first National Disaster Management Plan

2. Odisha super cyclone

3. Disaster Management Act passed

4. Disaster Management (Amendment) Act passed

Select the correct order:

A. 3, 2, 1, 4

B. 2, 1, 3, 4

C. 2, 3, 1, 4

D. 2, 3, 4, 1

Answer: C. Super cyclone 1999 → DM Act 2005 → NDMP 2016 → Amendment Act 2025.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write hazard vs disaster, the risk formula and India's exposure figures. 2. Write each hazard with cause, regions, examples and warning agency. 3. Write the DM cycle, the DM Act and 2025 amendment, funds, Sendai and AP's experience.

ECONOMY · Fiscal policy: fiscal imbalance, deficit finance, FRBM

UNDERSTAND IT FIRST

What is it? Fiscal policy is the Government's use of **taxes, spending and borrowing** to steer the economy (from Latin *fiscus*, 'treasury'). It is run by the **Finance Ministry** through the **Budget**, while **monetary policy** (Day 34) is run by the RBI. When the Government spends more than it earns, it has a **deficit** and must **borrow**; too much borrowing raises interest costs, can push up prices and leaves less money for the private sector. The **FRBM Act** sets rules to keep this in check. Learn it as **what fiscal policy does → receipts and expenditure → kinds of deficit → deficit financing and its effects → FRBM and its journey → latest numbers → States**.

1. Tools, aims and the Budget's parts

- **Aims:** growth, **price stability**, **employment**, **reducing inequality** (progressive taxes, welfare spending), balanced regional development. **Expansionary (loose) fiscal policy** = more spending and/or lower taxes, used in a **slowdown** (Keynes: government spending boosts demand; the **multiplier** effect). **Contractionary (tight)** = less spending and/or higher taxes, used when the economy **overheats** or debt is too high. **Counter-cyclical** policy leans against the cycle; **automatic stabilisers** (tax revenue falls and welfare spending rises by itself in a slump) help.
- **Budget (Article 112, 'Annual Financial Statement')** has a **revenue account** and a **capital account**. **Revenue receipts** do not create a liability or reduce assets: **tax revenue** (income tax, corporation tax, GST, customs) and **non-tax revenue** (RBI dividend, interest, fees). **Capital receipts** either create a **liability** (**borrowings**, small savings) or reduce **assets** (**disinvestment**, recovery of loans). **Revenue expenditure** does not create assets: **interest payments** (the biggest item), salaries, pensions, **subsidies**, grants. **Capital expenditure** creates **assets** or reduces liabilities: **roads, railways, defence equipment**, loans to States, repayment of debt. Since about 2020 the Centre has sharply raised **capital expenditure** (about **₹11–12 lakh crore**, over **3% of GDP**) because it has a higher multiplier.

2. Kinds of deficit and deficit financing

Kinds of deficit

- The **fiscal deficit** is the most watched: it equals the Government's **total borrowing** in a year. A high **revenue deficit** is worse than a high capital deficit, because it means borrowing to pay **day-to-day** bills (salaries, interest) rather than to build assets; the **'golden rule'** says borrow only for investment. A **primary deficit** near zero means new borrowing is only paying **interest on old debt**. **Fiscal imbalance** also has a

vertical side (Centre collects more, States spend more) and a **horizontal** side (rich vs poor States), handled by the **Finance Commission** (Day 32).

- **Deficit financing** means covering a deficit by **borrowing**, especially by **creating new money** (borrowing from the **central bank**). In India's early plans it funded development but also fed **inflation**. Until **1997**, the Government issued **ad hoc treasury bills** that the RBI automatically bought (automatic monetisation); an **RBI–Government agreement (1994)** ended this from **April 1997**, replacing it with limited **Ways and Means Advances**. The **FRBM Act** later **barred the RBI from buying government securities in the primary market** (except in special cases such as the **escape clause**). Effects of large deficits: **inflation**, higher **interest rates**, **crowding out** of private investment (Government borrowing absorbs savings), a growing **interest burden**, and a possible **twin deficit** (fiscal plus current account).

3. FRBM and the numbers

Fiscal rules in India

Stage	What happened
FRBM Act 2003	**Fiscal Responsibility and Budget Management Act** (rules 2004; based on the **E.A.S. Sarma** committee's work): bring the **fiscal deficit to 3% of GDP** and **eliminate the revenue deficit** by **2008–09**; three statements with the Budget (**Medium-Term Fiscal Policy**, **Fiscal Policy Strategy**, **Macro-economic Framework**); stop RBI primary purchases. States passed their own **Fiscal Responsibility Acts** (AP in **2005**).
Review and amendment	The **N.K. Singh FRBM Review Committee (report 2017)** made **debt** the main anchor: general government debt **60% of GDP** (Centre **40%**, States **20%**) by 2023, fiscal deficit **3%**, and an **escape clause** (up to **0.5%** of GDP extra for national security, war, calamity, collapse of agriculture or sharp fall in growth). The **2018 amendment** adopted much of this.
COVID and after	The Centre's fiscal deficit jumped to about **9.2% of GDP (2020–21)**. A **glide path** brought it down to **4.8% (2024–25)** and **4.4% (2025–26, revised)**, meeting the promise of **below 4.5%**. **Budget 2026–27** (1 February 2026) targets a fiscal deficit of **4.3%** and Central debt of about **55.6% of GDP**, with a new anchor of **debt at 50 ± 1% of GDP by 2030–31**. **States** may run fiscal deficits of **3% of GSDP** (extra **0.5%** linked to power-sector reforms), and get **50-year interest-free capex loans** from the Centre.

OTHER ANGLES ASKED IN EXAMS

Other angles asked: (1) **Fiscal deficit = borrowing**, so 'total borrowing and other liabilities' is the standard Budget definition. (2) **Interest payments** take about **two-fifths** of the Centre's **revenue receipts**. (3) **Disinvestment** is a **non-debt capital receipt**, so it reduces the fiscal deficit. (4) **Off-budget borrowing** (through public sector bodies) hides the true deficit; the Centre now discloses it. (5) The **Fiscal Council** idea (an independent watchdog) was suggested by the N.K. Singh and 14th/15th Finance Commissions but not created. (6) **Outcome Budget** (since 2005) and **Gender Budget** (since 2005–06) are other budgeting tools. (7) **Zero-base budgeting** starts every item from zero each year.

EXAM TRAP

Fiscal policy = taxes, spending, borrowing (Finance Ministry); aims: growth, stability, jobs, equity; **expansionary in slowdown, contractionary in boom**; **Keynes, multiplier, automatic stabilisers**; **Art 112**; **revenue receipts (tax, non-tax)**; **capital receipts (borrowings, disinvestment, loan recovery)**; **revenue expenditure (interest biggest, salaries, subsidies)**; **capital expenditure (assets, loans, debt repayment; ~₹11–12 lakh cr, >3% GDP)**; **RD = rev exp − rev receipts**; **ERD = RD − capital-asset grants**; **FD = total exp − (rev receipts + non-debt capital receipts) = borrowing**; **PD = FD − interest**; **monetised deficit**; **golden rule**; **ad hoc T-bills ended April 1997 (1994 agreement) → WMA**; effects: inflation, interest, **crowding out**, twin deficit; **FRBM 2003 (rules 2004): FD 3%, RD 0 by 2008–09; three statements; no RBI primary purchase**; **AP FRBM 2005**; **N.K. Singh 2017: debt 60% (40 + 20), escape clause 0.5%**; **2018 amendment**; **FD ~9.2% (2020–21) → 4.8% (2024–25) → 4.4% (2025–26 RE) → 4.3% (2026–27 BE)**; **debt ~55.6% (2026–27); 50 ± 1% by 2030–31**; **States 3% GSDP (+0.5% power)**; **50-year interest-free capex loans**.

Test yourself: 8 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Fiscal deficit is total expenditure minus revenue receipts and non-debt capital receipts.

2. Primary deficit is fiscal deficit minus interest payments.

3. Revenue deficit is revenue expenditure minus revenue receipts.

4. Effective revenue deficit is revenue deficit plus grants for creation of capital assets.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Effective revenue deficit = revenue deficit minus grants for creation of capital assets.

PRACTICE

Match List-I (event) with List-II (year):

List-I (event)	List-II (year)
A. FRBM Act enacted	1. 1997
B. N.K. Singh FRBM Review Committee report	2. 2030–31
C. End of ad hoc treasury bills	3. 2003

D. Target year for debt at 50 ± 1% of GDP	**4.** 2017

(1) A → 4 B → 3 C → 1 D → 2

(2) A → 3 B → 1 C → 4 D → 2

(3) A → 3 B → 4 C → 1 D → 2

(4) A → 3 B → 4 C → 2 D → 1

Answer: (3) FRBM 2003; N.K. Singh 2017; ad hoc T-bills ended 1997; debt anchor 2030–31.

PRACTICE

Which one of the following statements is NOT correct?

A. A high revenue deficit means borrowing to meet day-to-day expenses.

B. Disinvestment proceeds are non-debt capital receipts.

C. Interest payments are revenue expenditure.

D. A primary deficit of zero means the Government borrows nothing that year.

Answer: D. Zero primary deficit means borrowing equals interest payments; the Government still borrows to pay interest on old debt.

PRACTICE

Assertion (A): The Union Budget 2026–27 set a fiscal deficit target of 4.3% of GDP.
Reason (R): The FRBM Act was enacted in 2003.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but the 2003 Act's date does not explain the 2026–27 target, which follows the glide path and the new debt anchor.

PRACTICE

In a year, the Government's revenue receipts are ₹30 lakh crore, non-debt capital receipts ₹1 lakh crore, total expenditure ₹51 lakh crore and interest payments ₹13 lakh crore. What is the primary deficit?

A. ₹20 lakh crore

B. ₹13 lakh crore

C. ₹7 lakh crore

D. ₹33 lakh crore

Answer: C. Fiscal deficit = 51 − (30 + 1) = 20; primary deficit = 20 − 13 = ₹7 lakh crore.

PRACTICE

Consider the following statements:
I. An expansionary fiscal policy (more spending or lower taxes) is generally used during a slowdown.
II. 'Crowding out' means private investment rises when the Government borrows more.
Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: A. Crowding out means heavy Government borrowing reduces private investment by absorbing savings and raising interest rates.

PRACTICE

Consider the following pairs (item – Budget head):
1. Income tax collected – revenue receipt
2. Proceeds from selling shares of a public sector company – capital receipt
3. Building a national highway – revenue expenditure
4. Interest paid on public debt – capital expenditure
How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: B. Pairs 1 and 2 are correct. A highway is capital expenditure (it creates an asset); interest is revenue expenditure.

PRACTICE

Prices are rising fast and the economy is overheating. Which of the following is a fiscal (not monetary) step the Government can take?

A. Cut income tax rates

B. Raise subsidies on consumer goods

C. Raise the repo rate

D. Cut non-essential spending and reduce the fiscal deficit

Answer: D. Cutting spending and the deficit is contractionary fiscal policy; raising the repo rate is monetary policy (RBI); the other two add to demand.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the meaning, aims and kinds of fiscal policy, and the four Budget heads with examples. 2. Write RD, ERD, FD, PD and monetised deficit with formulas, deficit financing and its effects. 3. Write the FRBM journey (2003, 2017, 2018, COVID, glide path, debt anchor) and the latest numbers for the Centre and States.

SCIENCE · General science in everyday life: physics, chemistry, biology

UNDERSTAND IT FIRST

What is it? APPSC asks **everyday science**: why the sky is blue, why ice floats, what baking soda is, why a vitamin shortage causes a disease. These questions look easy but carry **traps** in units, formulas and pairs. This page collects the most-asked basics in three parts. Learn it as **physics (units, motion, light, sound, electricity) → chemistry (acids and bases, common compounds, metals) → biology (cells, vitamins, blood, body facts)**, and always link a fact to an **everyday example**.

1. Physics

Topic	Key facts with everyday examples
Units (SI)	**Seven base units**: metre (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (amount), candela (light intensity). Derived: **newton** (force), **joule** (work, energy), **watt** (power), **pascal** (pressure), **hertz** (frequency), **ohm** (resistance), **volt**, **coulomb**. **1 kWh** = 1 'unit' of electricity on your bill. **Light year** is a unit of **distance**.
Motion and force	**Newton's 1st law (inertia)**: passengers jerk forward when a bus brakes. **2nd law: F = ma**: a cricketer pulls his hands back to catch (more time, less force). **3rd law (action–reaction)**: rockets, swimming, a gun's recoil. **Gravity**: g ≈ **9.8 m/s²**; weight on the Moon is about **one-sixth**; mass stays the same everywhere. **Archimedes' principle**: ice floats because it is **less dense** than water; ships float by displacing water.
Sound and heat	Sound needs a **medium** (no sound in space); fastest in **solids**, then liquids, slowest in gases (about **343 m/s** in air). **Echo** needs at least about **17 m** distance; **ultrasound** (above 20,000 Hz) used in scans and by bats; **infrasound** (below 20 Hz) by elephants. **Decibel** measures loudness. Heat moves by **conduction** (metal spoon), **convection** (sea breeze, boiling water) and **radiation** (sunlight; no medium needed). Water is densest at **4°C**, so lakes freeze from the top.
Electricity and magnetism	**Ohm's law: V = IR**. Household supply in India: **220–230 V, 50 Hz AC**; home wiring is in **parallel** (each appliance gets full voltage). A **fuse** (low melting point wire) or **MCB** breaks the circuit on overload; **earthing** protects from shocks. **Electric motor**: electrical → mechanical energy; **generator (dynamo)**: mechanical → electrical (Faraday's **electromagnetic induction**). **Transformers** work only on **AC**. **LED** bulbs save energy.



- **More on light:** light travels fastest in a **vacuum** (about **3 lakh km/s**) and slows in water and glass, which is why it **bends**. **Convex lens** (converging) for **long-sightedness (hypermetropia)**, magnifying glasses; **concave lens** (diverging) for **short-sightedness (myopia)**. **Concave mirrors**: torches, shaving and dentists' mirrors; **convex mirrors**: vehicle rear-view mirrors (wider view). **Primary colours of light** are red, green, blue. The **danger signal is red** because red light scatters least and is seen from far.

2. Chemistry

Topic	Key facts
Acids, bases, pH	**pH scale 0–14**: below 7 **acidic**, 7 **neutral** (pure water), above 7 **basic**. Acids turn **blue litmus red**; bases turn **red litmus blue**. Natural acids: **citric** (lemon), **acetic** (vinegar), **lactic** (curd), **tartaric** (tamarind, grapes), **formic** (ant sting), **oxalic** (spinach, tomato); **hydrochloric acid** in the stomach (pH about 1.5–2). Bases: **milk of magnesia** (magnesium hydroxide) as an **antacid**; **baking soda** on a bee sting (acidic); **vinegar** on a wasp sting (alkaline). Healthy blood pH about **7.4**; **acid rain** below **5.6**; tooth decay starts below about **5.5**.
Common compounds	**Common salt** NaCl; **baking soda** $NaHCO_3$ (sodium bicarbonate; cakes, antacid, fire extinguishers); **washing soda** $Na_2CO_3 \cdot 10H_2O$ (removes **permanent hardness** of water); **bleaching powder** $CaOCl_2$

	(disinfects drinking water); **plaster of Paris** $CaSO_4 \cdot \frac{1}{2}H_2O$ (from **gypsum** $CaSO_4 \cdot 2H_2O$; sets hard with water; fracture casts, statues); **quicklime** CaO, **slaked lime** $Ca(OH)_2$ (whitewash); **dry ice** = solid CO_2; **laughing gas** = N_2O; **heavy water** D_2O (moderator in nuclear reactors); **marsh gas** = methane; **LPG** mainly **butane and propane** (a smelly **ethyl mercaptan** is added to detect leaks); **CNG** mainly **methane**.
Metals, alloys, rusting	**Mercury** is the only metal **liquid** at room temperature (bromine is the liquid non-metal); **silver** is the best conductor, **copper** is used for wires; **tungsten** (highest melting point) in old bulb filaments; **aluminium** is the most abundant metal in the crust (from **bauxite**). **Alloys: brass** = copper + zinc; **bronze** = copper + tin; **stainless steel** = iron + chromium (+ nickel, carbon); **solder** = lead + tin; **German silver** = copper + zinc + nickel (no silver!); **amalgam** = mercury + another metal. **Rusting** needs **both water and oxygen**; prevented by paint, oil, **galvanisation** (zinc coating), alloying. **Hard water** has calcium and magnesium salts; **temporary** hardness (bicarbonates) goes on **boiling**.

3. Biology

Topic	Key facts
Cells	The cell is the **basic unit of life** (Schleiden and Schwann, cell theory). **Mitochondria** = 'powerhouse' (energy as ATP); **ribosomes** make proteins; **nucleus** holds DNA; **lysosomes** = 'suicide bags'; **chloroplasts** (plants) do photosynthesis; plant cells have a **cell wall** (cellulose) and a large vacuole. **Prokaryotes** (bacteria) have no true nucleus. **DNA** double helix (Watson and Crick, 1953); humans have **46 chromosomes (23 pairs)**; sex decided by the **father's X or Y**.
Vitamins and minerals	**Fat-soluble: A, D, E, K**; water-soluble: **B group, C**. **A** (retinol) → **night blindness**, xerophthalmia; **B1** (thiamine) → **beriberi**; **B3** (niacin) → **pellagra**; **B12** → pernicious **anaemia** (found in animal foods); **C** (ascorbic acid) → **scurvy** (bleeding gums); **D** (made in skin with sunlight) → **rickets** in children, osteomalacia in adults; **E** → fertility, antioxidant; **K** → **blood clotting**. Minerals: **iron** → anaemia; **iodine** → **goitre** (iodised salt); **calcium** → bones; **fluoride** in excess → **fluorosis** (Prakasam, Nalgonda belt).
Blood and body	**Blood groups** (Landsteiner): A, B, AB, O by **antigens** on red cells; **O negative** = **universal donor** (no A or B antigen), **AB positive** = **universal recipient** (no anti-A or anti-B antibodies); **Rh** factor matters in pregnancy. **RBCs** carry oxygen with **haemoglobin** (iron), live about **120 days**, made in **bone marrow**; **WBCs** fight infection; **platelets** help clotting (fall in **dengue**). Adult has **206 bones**; smallest bone **stapes** (ear), longest **femur** (thigh); largest gland **liver** (makes **bile**); **insulin** from the **pancreas** controls blood sugar; normal BP about **120/80 mm Hg**; heart has **4 chambers**; **kidneys** filter blood (nephrons).

OTHER ANGLES ASKED IN EXAMS

Other angles asked: (1) **Mass vs weight**: mass in kg never changes; weight (a force, in newtons) changes with gravity. (2) **Pressure cookers** cook faster because higher pressure **raises the boiling point**; at high altitude water boils **below 100°C**. (3) **Stars twinkle** (refraction through moving air layers) but **planets do not** (they are discs, not points). (4) **German silver has no silver**; **lead pencils have no lead** (graphite). (5)

Photosynthesis releases oxygen that comes from **water**, not from CO_2. (6) **Antibiotics do not act on viruses** (Day 35 link: dengue, COVID). (7) **Kelvin = °C + 273**; **−40** is the same in °C and °F.

EXAM TRAP

7 SI base units; **newton, joule, watt, pascal, hertz, ohm**; **1 kWh = 1 unit**; **light year = distance**; **Newton I (inertia: bus braking), II (F = ma: catching), III (rocket, recoil)**; **g 9.8; Moon 1/6**; **Archimedes: ice floats**; sound: **medium needed; solids fastest; ~343 m/s air; echo ≥17 m; ultrasound >20 kHz (bats), infrasound <20 Hz (elephants); decibel**; **conduction, convection, radiation**; **water densest at 4°C**; **V = IR; 230 V, 50 Hz; parallel wiring; fuse, MCB, earthing; motor vs generator (Faraday); transformer AC only**; light: **reflection, refraction (pool, twinkling), TIR (fibre, diamond, mirage), dispersion (VIBGYOR; violet bends most), scattering (blue sky, red danger)**; **vacuum fastest (3 lakh km/s)**; **convex lens: hypermetropia; concave: myopia; convex mirror: rear-view**; **pH 0–14; litmus**; acids: **citric, acetic, lactic, tartaric, formic, oxalic; stomach HCl**; **antacid $Mg(OH)_2$**; **blood 7.4; acid rain <5.6; teeth <5.5**; **$NaHCO_3$ baking soda**; **$Na_2CO_3 \cdot 10H_2O$ washing soda**; **$CaOCl_2$ bleaching**; **$CaSO_4 \cdot \frac{1}{2}H_2O$ PoP (from gypsum)**; **CaO, $Ca(OH)_2$**; **dry ice CO_2**; **N_2O laughing**; **D_2O heavy water**; **LPG butane + propane (mercaptan)**; **CNG methane**; **mercury liquid metal; silver best conductor; tungsten; aluminium most abundant (bauxite)**; **brass Cu + Zn; bronze Cu + Sn; stainless steel Fe + Cr (+ Ni); solder Pb + Sn; German silver Cu + Zn + Ni**; **rust: water + oxygen; galvanisation (zinc)**; **hard water; temporary goes on boiling**; **mitochondria, ribosomes, lysosomes, chloroplasts; 46 chromosomes; DNA 1953**; vitamins: **A night blindness, B1 beriberi, B3 pellagra, B12 anaemia, C scurvy, D rickets, K clotting; ADEK fat-soluble**; **iodine goitre; fluoride fluorosis**; **O− universal donor, AB+ universal recipient; RBC 120 days, bone marrow; 206 bones; stapes, femur; liver largest gland; insulin pancreas; BP 120/80**.

Test yourself: 8 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The sky looks blue because blue light is scattered more by air molecules.

2. A rainbow is formed by refraction, dispersion and internal reflection of sunlight in raindrops.

3. Sound travels faster in solids than in air.

4. Light travels faster in water than in a vacuum.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. Light is fastest in a vacuum and slows down in water and glass.

PRACTICE

Match List-I (common name) with List-II (formula):

List-I (common name)	List-II (formula)
A. Baking soda	1. $CaOCl_2$
B. Washing soda	2. $CaSO_4 \cdot \frac{1}{2}H_2O$
C. Plaster of Paris	3. $Na_2CO_3 \cdot 10H_2O$
D. Bleaching powder	4. $NaHCO_3$

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Baking soda $NaHCO_3$; washing soda $Na_2CO_3 \cdot 10H_2O$; PoP $CaSO_4 \cdot \frac{1}{2}H_2O$; bleaching powder $CaOCl_2$.

PRACTICE

Which one of the following statements is NOT correct?

A. Deficiency of vitamin A causes night blindness.

B. Deficiency of vitamin C causes rickets.

C. Vitamin K helps blood to clot.

D. Vitamin D is made in the skin in sunlight.

Answer: B. Vitamin C deficiency causes scurvy; rickets is due to vitamin D deficiency.

PRACTICE

Assertion (A): A person with blood group O can safely receive blood from a donor of group AB.
Reason (R): Red blood cells of group O have neither A nor B antigens.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: a group O person has anti-A and anti-B antibodies and can receive only group O blood. R is true, which is why O is the universal donor.

PRACTICE

A swimming pool looks shallower than it really is, and a straight stick dipped in it looks bent. Which phenomenon explains both?

A. Refraction of light

B. Reflection of light

C. Diffraction of light

D. Scattering of light

Answer: A. Light bends (refracts) as it passes from water to air, so objects under water appear raised.

PRACTICE

Consider the following statements:
I. Stainless steel is an alloy of iron with chromium (and usually nickel).
II. Brass is an alloy of copper and zinc.
Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: C. Both are correct; bronze (copper + tin) is the alloy often confused with brass.

PRACTICE

Consider the following pairs (quantity – SI unit):

1. Force – newton

2. Power – watt

3. Electric resistance – ohm

4. Frequency – hertz

How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: D. All four units are correct.

PRACTICE

A person suffering from acidity (excess stomach acid) is given an antacid. Which substance does a common antacid like 'milk of magnesia' contain?

A. Hydrochloric acid

B. Magnesium hydroxide

C. Sodium chloride

D. Acetic acid

Answer: B. Magnesium hydroxide is a mild base that neutralises excess stomach acid.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write SI units, Newton's laws, sound, heat and electricity with one example each, and the five light effects. 2. Write pH with examples, the common compounds with formulas and uses, and metals, alloys, rusting and hard water. 3. Write cell parts, vitamins and minerals with deficiency diseases, blood groups and key body facts.

NOTEBOOK PAGE · Day 37: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Disasters and their management

Area	Write
Basics	Hazard vs disaster; risk = hazard × vulnerability ÷ capacity; India's exposure figures; each hazard (cyclone, flood, drought, earthquake, tsunami, landslide, heatwave, lightning, urban flood) with cause, regions and warning agency
Management	DM cycle; DM Act 2005 and 2025 amendment (NDMA, SDMA, DDMA, UDMA, NDRF, SDRF, NIDM, funds); NPDM 2009, NDMP 2016; Hyogo, Sendai (4 priorities, 7 targets); AP's disasters and APSDMA

ECONOMY: Fiscal policy and FRBM

- Meaning and tools of fiscal policy; expansionary vs contractionary; revenue vs capital receipts and expenditure; RD, ERD, FD, PD, monetised deficit with formulas; deficit financing and its effects; crowding out; FRBM 2003, N.K. Singh 2017, 2018 amendment, escape clause, glide path, debt anchor; latest numbers; States' limits.

SCIENCE: Everyday science

Write: SI units; Newton's laws with examples; light (five effects with examples), sound, electricity basics; acids, bases, pH and examples; common compounds with formulas and uses; alloys and rusting; cells; vitamins and deficiency diseases; blood groups and donation rule; key body facts.